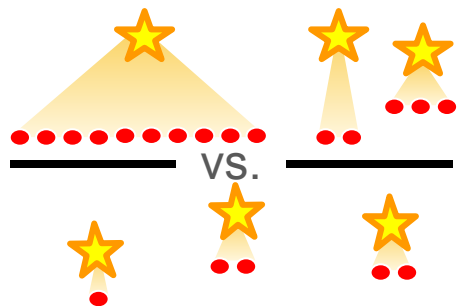




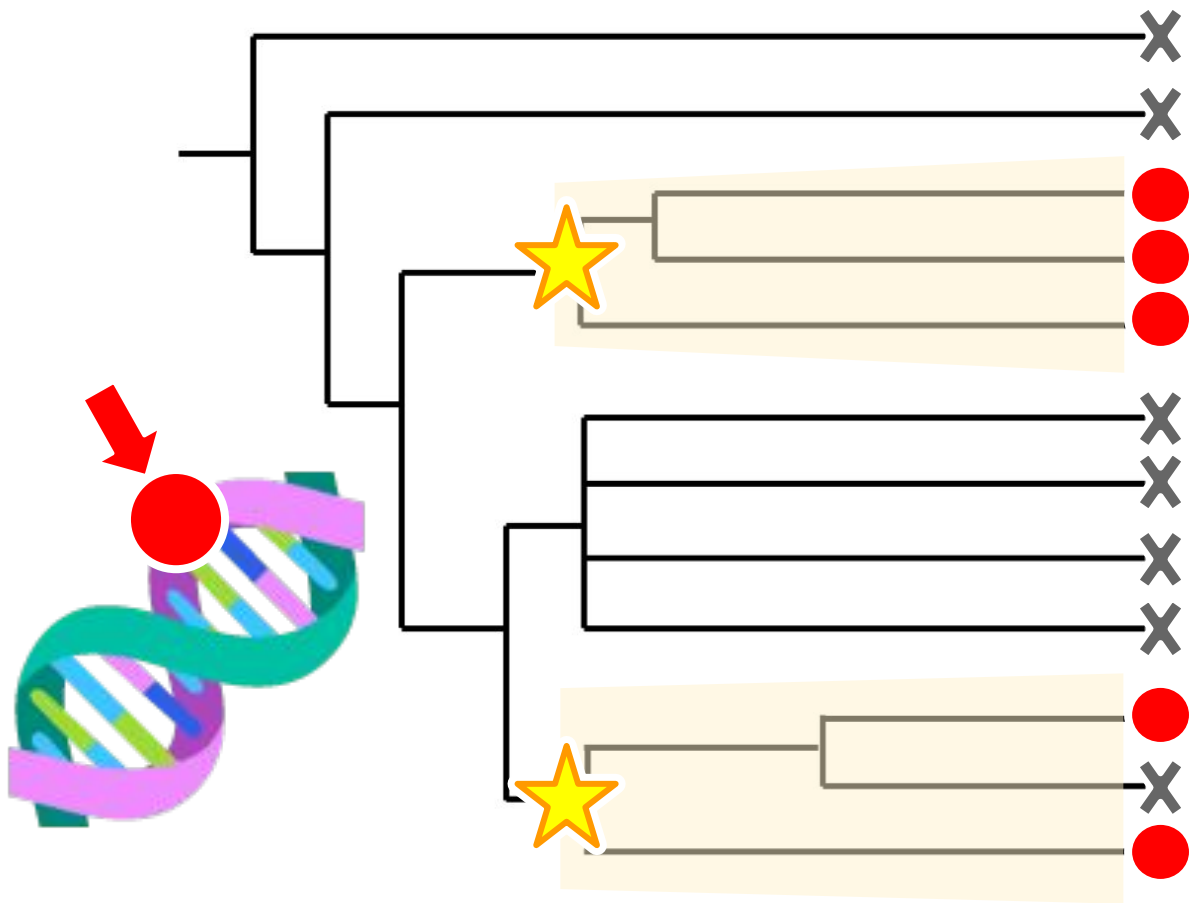
vs.







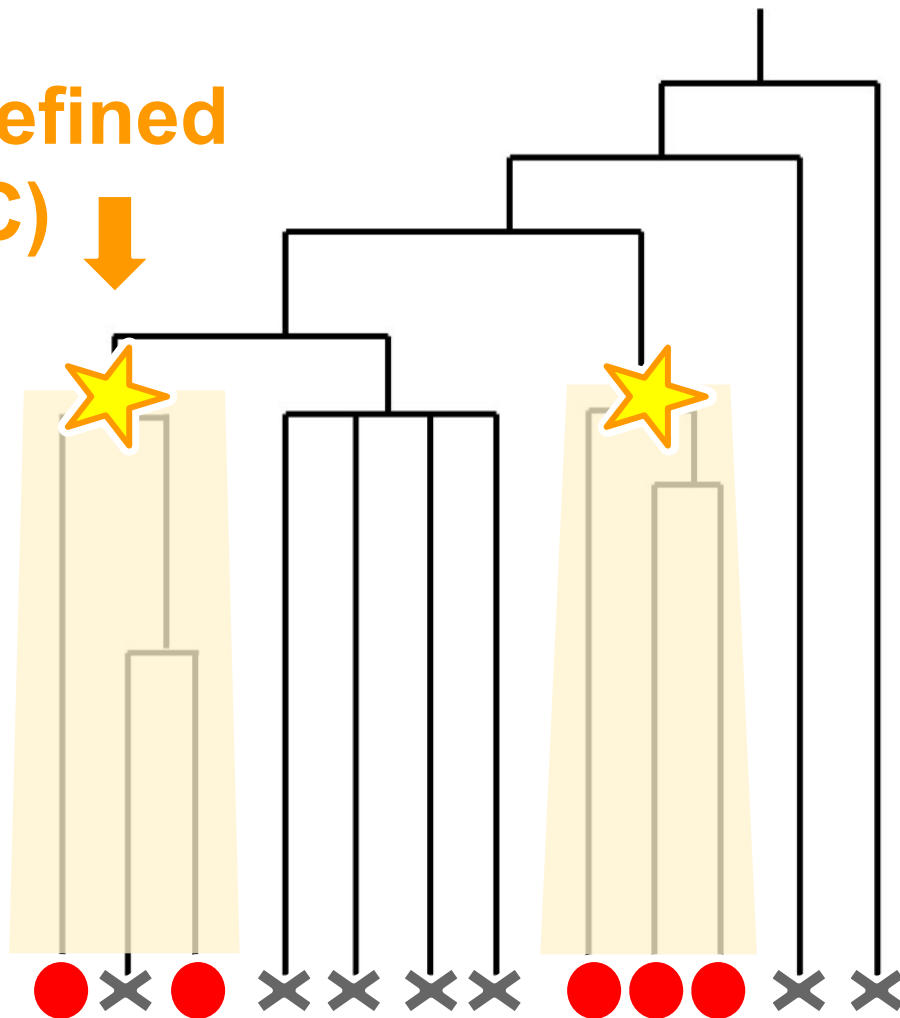




**Mutation-Defined
Clade (MDC)**



**Mutation
of Interest**



*within-host
model* / *covasim
ABM*

*transmission
tree*

*pathogen
phylogeny*

*sequence
generation*

